

Chapter 18

Coatings for Prevention or Deactivation of Biological Contaminants

Joerg C. Tillier

*Department of Bio- and Chemical Engineering, Technical University Dortmund,
Dortmund, Germany*

Chapter Outline

18.1 Introduction	752		
18.2 Biological Contaminations	754		
18.3 Means of Contamination	754		
18.4 General Requirements for Self-Cleaning Coatings	755		
18.5 Laboratory Tests for Antimicrobial Activity of Coatings	755		
18.6 Agents Against Biological Contaminations	757		
18.7 Coating Methods	757		
18.7.1 Brush, Pad, and Roll Coating	758		
18.7.2 Dip and Flow Coatings	758		
18.7.3 Spin Coating	758		
18.7.4 Spray Application	758		
18.7.5 Electroplating	758		
18.7.6 Electroless Plating	759		
18.7.7 Sputtering	759		
18.7.8 Physical Vapor Deposition	759		
18.7.9 Chemical Vapor Deposition	759		
		18.7.10 Surface Modification	759
		18.8 Non-Adhesive Coatings	760
		18.8.1 Coatings with Hydrophilic Polymers and Hydrogels	760
		18.8.2 Ultrahydrophobic Coatings	762
		18.8.3 Influence of Surface Net Charge on Microbial Adhesion	763
		18.8.4 Proteins	763
		18.9 Microbe Killing or Growth Inhibiting Coatings	764
		18.9.1 Release Systems	764
		18.9.1.1 Diffusion Systems	764
		18.9.2 Contact Active Antimicrobial Surfaces	772
		18.9.2.1 Chemical Action	774
		18.9.2.2 Physicochemical Action	774

18.10 Metal Coatings	776	18.15.1 Biomedical Applications	780
18.11 Antifouling Paints	777	18.15.2 Food Protection	783
18.12 Antimicrobial Surfaces with Multiple Action	779	18.15.3 Textiles	783
18.13 Antiviral Coatings	779	18.15.4 Daily Life Products	783
18.14 Surface Cleaning by Coating	779	18.15.5 Construction and Ships	784
18.15 Application Examples	780	18.16 Future Developments	784
		References	784

18.1 INTRODUCTION

Materials in natural and industrial aquatic ecosystems as well as in any populated natural environment will become biologically contaminated sooner or later. In the context of this chapter a biological contamination is a surface impurity consisting of viable organisms. These contaminants can result in numerous highly undesirable events, such as mold growth in kitchens and bathrooms; fungal overgrowth of house walls; poisoning of flowing systems, e.g., water pipes or tubing of vending machines; biofilm growth on ship hulls; nosocomial infections on biomedical devices, e.g., catheters or implants; and virulent infections contact-transmitted by frequently touched surfaces in daily life.

The first step of such a contamination is most often the attachment of microbes. Micro-organisms may cause this type of primary infection from either airborne or aqueous vehicles. Another mode of transmission occurs by the material coming in contact with contaminated soil, invertebrates, plants, animals or humans. Depending on the nature of the contaminating micro-organism the surfaces may become instantly able to transmit virulent infections on contact. This is particularly the case for frequently touched items of daily life, such as door knobs, computer keyboards, and telephone receivers. After their first contact, the microbes might under humid conditions proliferate on the surface and eventually form a biofilm, a layer composed of micro- and eventually macro-organisms embedded in a biopolymer matrix. The overgrowth of a material with a biofilm is referred to as biofouling. This process occurs in a wide variety of environments and leads to infections, toxification, and deterioration of materials (Figure 18.1).

The majority of biofouling processes is based on attached micro-organisms, such as bacteria, yeast, fungi, and micro-algae, but also macro-organisms, e.g., invertebrates, clams or phytoplankton, can be involved in biofilm formation particularly in maritime systems. Many plants and animals have their own natural defense systems against microbial attachment, such as production of toxins or repellants,^{1,2} or a continuous sloughing off of surface layers.³ Man-made materials, however, usually do not have this innate ability to defend themselves against biological attacks.



FIGURE 18.1 Typical biofilm contaminations: (a) fungus at wall, (b) algae at ship hull, (c) bacteria at catheter.

A common way to prevent events caused by biological contamination of materials is to frequently clean their surfaces. This can be realized by material specific techniques, e.g., with surfactants and ultrasound (described in other chapters of this book). In a number of cases it would also be sufficient to use disinfectants, such as sodium hypochloride or 70% ethanol. Yet in practice neither of the methods can really prevent events caused by biological contamination. Further, the use of cleaning agents other than water has some environmental problems; disinfectants additionally support the formation of resistant microorganisms, which is a serious problem in modern medicine. Therefore, a better method of defense than simply cleaning surfaces would be to alter them in such a way that no biological contamination is able to occur in the first place. Such biocidal surfaces can be called self-cleaning because of their ability to continuously defend the coated material against biological attacks. Since functionality and durability of products are in danger when biological contamination occurs, and since it is also a potential threat to human health, it is both economically and medically desirable to supply items with a biocidal layer. These methods may be applied to a wide range of products, including biomedical devices (e.g., catheters, contact lenses, and implants), contact surfaces of daily life (e.g., computer keyboards, door knobs, and even bank notes), and untouched surfaces (e.g., ship or house walls, drinking water pipes).

In recent decades numerous antimicrobial and antifouling coatings have been developed for almost every possible application. Their goal was always to achieve a surface that prevents biological contamination longer than the expected life span of the product, which could vary between a few hours and several years. A number of remarkably effective solutions have been designed. Yet most of the coatings exhibit unacceptable disadvantages regarding their material properties, life span, and level of environmental toxicity. The purpose of this chapter is to classify existing solutions for biological contamination-preventing coatings with respect to their functional principles, show application

examples, discuss their advantages and disadvantages, and provide a perspective on possible future developments.

18.2 BIOLOGICAL CONTAMINATIONS

Most biological contaminations are based on *microbes*, which is a general term for micro-organisms, such as bacteria, yeast, fungi, or algae (for a review see Ref. 4). *Bacteria* are mainly unicellular micro-organisms without a nucleus (the size is in a range of 1–5 μm). They multiply by means of simple division or in some cases through sprouting. Bacteria are divided into two sub-groups: Gram-positive (cytoplasmic membrane, cell wall 20–80 nm in thickness; representatives: *Staphylococcus aureus* and *Staphylococcus epidermidis*) and Gram-negative (cytoplasmic membrane, cell wall $\sim$ 6–10 nm in thickness, and outer membrane: representatives: *Escherichia coli* and *Pseudomonas aeruginosa*) strains. These groups often exhibit very different behaviors regarding adhesion to surfaces and susceptibility to antibiotics. Due to their thick cell walls, which consist of petidoglucans, bacteria can survive under extreme conditions (heat, cold, dryness, radiation, etc.) either as cells or in some cases as spores (best known for *Bacillus anthrax*). Due to their variable feeding methods, bacteria are ubiquitous and infections can therefore occur almost everywhere.

Yeast and Fungi are a heterogeneous group of micro-organisms with genuine nuclei. The cell walls of their nuclei consist of chitin and/or cellulose; the cell walls also consist of polysaccharides. The unicellular and multicellular forms multiply both sexually and asexually, in many different forms, particularly through the formation of flagellated and unflagellated spores, which are especially resistant against extreme conditions, such as heat, pressure, and even chemical treatment with bleach. Fungi (representatives: *Aspergillus niger*, *Penicillium notatum*, and *Stachybotrys chartarum*) and yeasts (representatives: *Candida albicans* and *Saccharomyces cerevisiae*) occur in all climatic zones of the earth, especially where there is high humidity. There are two forms: those developing on water and those developing on solid substrates. Particularly because of the high stability of the spores, which can easily be moved in water or through air (e.g., wind), contamination can occur everywhere. Mildew and mold growth are the most common outcomes of a fungal contamination in a moist environment.

Algae are a group of plants with vegetative bodies of one or more cells, containing nuclei. They occur in the sea, in freshwater, in humid places such as on wet walls, on the floor, and on snow.

Apart from microbes, *invertebrates*, *clams*, *barnacles*, and *mussels* are also involved in surface fouling, especially in maritime and limnic biological contaminations.

18.3 MEANS OF CONTAMINATION

Biological contamination can occur in three general ways: through air, in liquid or by contact. Micro-organisms can attach from the air, e.g., bacterial or fungal

spores, bacterial cells. Spores are especially stable on surfaces and can be transmitted by contact. The attached airborne cells can proliferate under humid conditions and form biofilms. Another mode of contamination is the attachment of waterborne organisms, which are ubiquitous in every moist environment. These organisms proliferate on a surface, which leads to biofilm formation, also referred to as biofouling. Third, micro-organisms can be attached to surfaces by contact with an already contaminated source. In this case the microorganisms are especially viable, because they are already in a matrix (lipids and proteins) that provides perfect living conditions.

18.4 GENERAL REQUIREMENTS FOR SELF-CLEANING COATINGS

The requirements for a biologically self-cleaning coating are strongly dependent on the application of the material. In general, they ought to be effective for all relevant contaminating organisms in the typical application environment of the material. Furthermore, the coating should be long lasting in its efficiency, it should not interfere with the necessary surface properties of the material, e.g., hardness or optical appearance, and it should be nontoxic to its environment. In general, there are two ways of retaining surfaces sterile as shown in [Figure 18.2](#). One is to repel microbes and the other is to kill them. In both cases, the microbes cannot form a biofilm, which is the most critical biological contamination. Surfaces can repel microbes if they are modified with a non-binding hydrophilic material, a negative charge, or are rendered ultrahydrophobic. Another way to repel microbes is coatings that slowly hydrolyze and regenerate themselves by frequently washing off the top layers. Killing microbes is usually realized by releasing biocides from surfaces that kill everything in the surrounding. Newer approaches deal with surfaces that kill microbes on contact.

18.5 LABORATORY TESTS FOR ANTIMICROBIAL ACTIVITY OF COATINGS

It takes considerable effort to test the efficacy of antimicrobial coatings under real conditions, because the number of germs is usually low and the probability

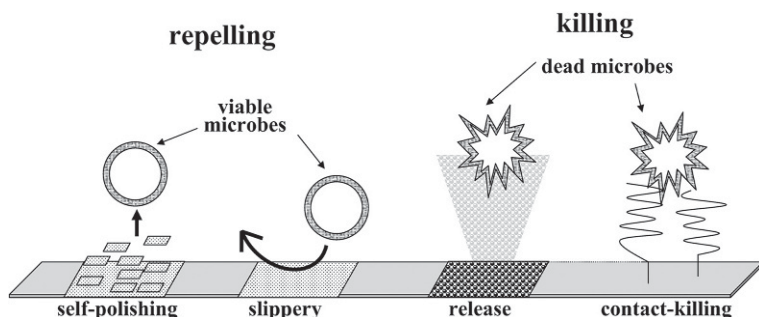


FIGURE 18.2 General approaches for antimicrobial surfaces.

of an infection or biofilm formation is often less than 1%, i.e., at least 1000 experiments should be done to obtain data that allow one to analyze the quality of the coating compared to an uncoated control sample. This is why antimicrobial coatings are usually tested with higher germ concentrations under simulated “real” conditions.

In order to measure the number of microbes adhered to surfaces, the samples (modified with microbe-repelling coatings) are placed into a suspension of microbial cells in water, phosphate-buffered saline (PBS), growth medium, plasma or serum and incubated stationary, under shaking, or flowing conditions (e.g., by using a Robbins device). After a certain incubation time, which can vary between 5 min and several days, the samples are rinsed, dried and the number of adhered microbes is most commonly counted using an inverted microscope, fluorescence microscope, or scanning electron microscope (SEM) (for a review see Ref. 5). In investigations of biofilm formation the samples with the adhered microbial cells are incubated at least over night in a growth medium (e.g., minimal medium) prior to taking cell counts.

The effectiveness of a microbe-killing coating may itself be tested by placing the coated sample into a microbial suspension, incubating it, and determining the reduction of the number of viable cells in suspension compared to a control. Even if this test is already used as industry standard (Dow suspension test), it does not give the entire information of the surface properties, because it provides no information about microbes adhered on the surface of the coated material. Alternatively, microbes are allowed to adhere on surfaces as described for microbe-repelling surfaces above. Then they are removed by washing with detergents (e.g., Tween) eventually assisted by ultrasound or by pressing solid agar on the surface. The most commonly used test at the moment is the so-called film adherence method.⁶ Here, a droplet of a bacterial suspension is placed onto the surface to be tested and covered with a defined piece of glass or plastic. After incubation for 1-48 h, the number of microbe in the whole sample is determined. A log-reduction of 3-4 compared to a control sample is considered sufficient to call a surface antimicrobial. In all cases the viability of the microbes is determined after having contact with the surface. One requirement for such a test is the total removal of all microbes from the surface without damaging them, which cannot be performed in some cases, e.g., in case of deposition of airborne vehicles. Further, bacteriostatic effects of such surfaces cannot be measured this way. Therefore, it is desirable to determine the viability of the microbes directly on the surface (note that microbes are often less susceptible against microbicides, when attached to surfaces).

One possibility of determining the killing efficiency of antimicrobial surfaces is the use of fluorescent stains, such as the live/dead kit, which reveals if the cytoplasmic membrane of a microbial cell is damaged or intact. Unfortunately, such staining methods were found to be inaccurate in several cases, depending on the kind of microbes and the nature and function of the investigated surface. The most reliable tests so far are those which explore the ability of surface-attached

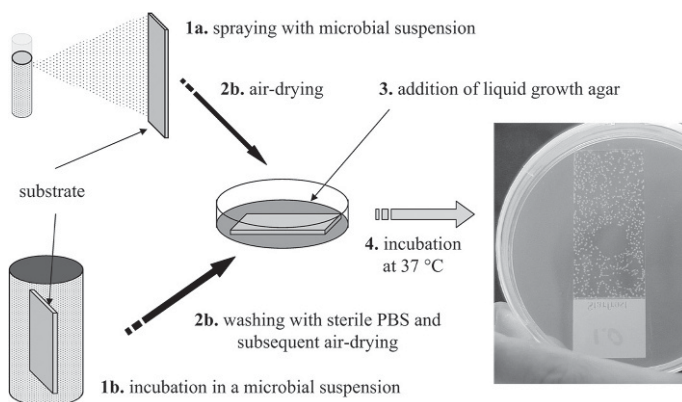


FIGURE 18.3 Colony test for antimicrobial properties of surfaces with microbes in direct contact.

microbes to proliferate, which is the most relevant property of viable microbes. A qualitative system for that is the agar-diffusion test, where the infected samples (either in microbial suspension or by spraying for instance spores) are cultivated on growth agar (for repeatability and reproducibility of germicide tests see Ref. 7). A recently developed qualitative colony test is based on the cultivation of infected samples under growth agar, allowing determination of the exact number of viable cells by colony count (Figure 18.3).^{8–10}

Materials that release biocide are commonly tested by immersing them into a microbial suspension and detecting the decrease of the number of viable cells in the medium. Another procedure is the agar-diffusion test, where the materials are contacted with growth agar and the whole system is inoculated with a microbial suspension. After cultivating overnight the inhibition zone (no growth of microbes) around the releasing material is measured.

18.6 AGENTS AGAINST BIOLOGICAL CONTAMINATIONS

The most general and therefore non-specific agents against germs are *biocides*, which kill all kinds of germs and also lower organisms, such as invertebrates or mussels. *Antimicrobial drugs* are specific killing microbes and can be classified as *microbiostatics*, which prevent proliferation (although not killing the cells), and *microbiocides*, which kill microbes. In case of more specific interactions these drugs are categorized further, e.g., as bacteriostats, bactericides, fungistats, fungicides, etc.

18.7 COATING METHODS

The major coating technologies, which are used to create antimicrobial surfaces so far, will be discussed shortly. For background reading with more detailed descriptions, the books by Paul,¹¹ Wicks et al.¹², and Satas and Tracton¹³ are recommended.

18.7.1 Brush, Pad, and Roll Coating

The classic coating is a manually or automatically controlled application of viscous, filmforming liquids to surfaces by brushes, pads, and rolls. Most commonly the coating is formed by air drying.

18.7.2 Dip and Flow Coatings

A material is immersed into a rather viscous solution or suspension of a coating-forming material, or into the single compound liquid [e.g., precursor of a polymer (monomer) or melted polymer] itself and then slowly removed in the same manner (dip) from the solution or through the solution (flow). The liquid remaining on the surface is then simply air dried, treated in vacuum or heated. A film is formed onto the surface by removing the solvent and/or polymerizing the monomer [e.g., by heat or UV (ultraviolet) radiation], or by just cooling the polymer melt.

18.7.3 Spin Coating

A liquid used for dip coating is applied to a rapidly rotating surface and a coating is formed in the same way. The advantage of spin coatings is the perfectly distributed coating material on the surface.

18.7.4 Spray Application

Another common way of coating surfaces is to simply spray a liquid formulation of the coating material onto a surface and subsequently allow the solvent to evaporate. Depending on the coating material and the substrate, the liquid can be applied under air, airless, with electric charge, as a temperature conditioned spray, or as a solution in supercritical CO₂. The spraying can be controlled by hand or as used in many industrial applications automated, e.g., by robots. In the case of powder coatings, the powdered coating material is sprayed at high temperatures followed by baking.

18.7.5 Electroplating

Metals as well as polyionic polymers can be applied by electroplating, also called electrodeposition, to conductive materials, such as aluminum and steel. The substrate is immersed into an aqueous solution of a metal salt or a polyion in an electrically conductive tank. The substrate is then cathodized for deposition of metals and positively charged polymers and anodized for negatively charged polymers. The desired film thickness can be controlled by the reaction time and the concentration of the solutions.

18.7.6 Electroless Plating

In order to coat a wide variety of materials, such as non-conductors (e.g., polymers), semiconductors or conductors with metal layers, a metal can be deposited from its salt solution by a chemical process. Here the substrate is placed into the metal salt solution and a reducing agent, e.g., hydrazine, is added. The formed elemental metal is then deposited onto almost every material surface, which does not react with the metal salt solution in water. Another possibility is to spray the mixture described above to a surface, obtaining a similar yet not as well defined metal layer.

18.7.7 Sputtering

The ionization of molecules in their gas phase enables them to be accelerated between two electrodes and therefore be directed toward a substrate, which can be any material that is stable under the applied conditions (vacuum, local heat). Since the coating is possible with single molecules, very thin, controlled and well-distributed films on surfaces can be obtained. Most of the coating materials used in sputtering are metals and metal oxides, but especially in the case of ion beam sputtering polymers can also be sputtered. Besides ion beam sputtering, the process can be performed directly on a current diode (only for conductive substrates), assisted by radio frequency, or in a magnetron. Plasma-assisted sputtering methods are also commonly used.

18.7.8 Physical Vapor Deposition

Metal coatings can be applied to surfaces by heating a metal target in high vacuum. A substrate in the proximity of the evaporated metal will be coated with a metal layer in a very controlled manner.

18.7.9 Chemical Vapor Deposition

Metals, semimetals and their oxides can be deposited onto surfaces via chemical alteration of a gaseous organic compound into solid particles. The conversion of the gaseous compound is achieved by thermal decomposition, pyrolysis, reduction, oxidation or hydrolysis. As opposed to sputtering, the deposition can often be carried out at normal pressure. Depending on the application chemical vapor depositions are assisted by plasma, flame, glow charge, or laser.

18.7.10 Surface Modification

The chemical modification of surfaces leads to introduction of a monolayer of new reactive residues, which often change the behavior of the material and are therefore referred to coatings as well. Almost every material can be modified by a variety of chemical reactions directly or assisted by plasma, flame, ultrasound

and others. Because of the often impracticable conditions, such as organic solvents, irritating reagents, and long reaction times, these modifications are rarely used in industry but are common in laboratories.

18.8 NON-ADHESIVE COATINGS

The concentration of possibly virulent micro-organisms in aqueous environments, e.g., in drinking water and body fluids, is mostly well under the infectious limit. If such organisms attach to a surface they can proliferate locally, form biofilms, and either produce pathogens in high concentrations and/or infect the surrounding ecosystem. Therefore, it is not always necessary to kill micro-organisms but rather simply to prevent their adhesion. Non-adhesive coatings must have none or very few binding sites for the respective micro-organisms. These kinds of coatings are mostly described for antibacterial applications but might be effective for numerous other micro-organisms. Usually coatings that are non-adhesive for micro-organisms are applied in an industrial process and not by the customer.

The three basic principles for microbe-repelling coatings are electrostatic repellence, extreme hydrophilic or hydrophobic character, and self-polishing surfaces. In many of these cases these effects are used simultaneously.

18.8.1 Coatings with Hydrophilic Polymers and Hydrogels

It was found early¹⁴ that surfaces coated with water-soluble, swellable polymers or with hydrogels (networks that form stable gels with water) show drastically lower microbial adhesion compared to uncoated surfaces.¹⁵ The mechanism of microbial repellence is not yet clear. The most common explanation is the free surface energy approach,¹⁶ whereby bacteria cannot be attached because an energy minimum is reached by the largest possible amount of absorbed water. It is further discussed that hydrogel coatings do not adhere microbes, because of their lower potential of adhering certain proteins (e.g., fibronectin¹⁷), which can promote microbial adhesion. Another explanation is that the major compound of a hydrogel is water. Therefore, the number of possible binding sites for microbes is drastically lower than on most surfaces, which gives the microbe less chance to adhere to the surface.

Usable polymers for repelling microbes on surfaces are synthetic polymers, such as poly(acrylic acid) (PAA), poly(2-hydroxyethyl methacrylic acid) (PHEMA), poly(methacrylic acid) (PMAA), poly(vinylalcohol) (PVA), poly(ethylene oxide) (PEG), and poly(*N*-vinylpyrrolidone) (PVP),^{18,19} poly(methylloxazoline) (PMOX), poly(*N,N*-dimethylacrylamide) (PDMAA) as well as natural ones, e.g., alginates, carrageen, karaya gum²⁰ (for structures see Figure 18.4). Mixtures of polymers, copolymers, and cross-linked systems, such as poly(*N*-vinyl-pyrrolidone-poly(urethane) (PVP-PU), are also used.¹⁹ Surfaces

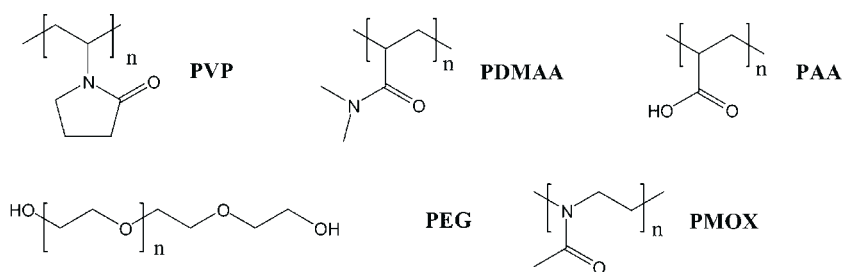


FIGURE 18.4 Typical microbe-repelling polymers: (a) PVP, (b) PDMAA, (c) PAA, (d) PEG, and (e) PMOX.

can be modified with these polymers either by simple coating techniques or by chemical attachment.

Coatings can adhere onto surfaces by ionic, hydrophilic/hydrophobic interactions. In order to achieve maximum adhesion the surfaces are activated or functionalized prior to coating. For instance, anionic polymers such as PMAA adhere better to surfaces modified with amino-groups, e.g., by silanization.^{18,21} These already commercially available coated materials prevent up to 95% microbial adhesion (e.g., tested for *S. aureus*, *S. epidermidis*, *E. coli*, and *Candida albicans*) in aqueous systems compared to the respective untreated control samples.²² For instance, the Hydromer[®]-coating, which consists of a mixture of poly (N-vinyl pyrrolidone) and several polyurethanes, applied to silicone prevented 92% of the adhesion of *S. epidermidis* compared to the untreated silicone.¹⁹ Coating of denture acrylic with sodium alginate led to a reduction of *Streptococcus salivarius*, a plaque causing bacterium, by 99% in vitro and still by 84% in vivo.²⁰ Another way of creating bacteria-repelling surfaces is the O₂ plasma treatment of plastics. It could be shown that such as treatment of poly(ethylene terephthalate) (PET) significantly lowers the adhesion of *S. epidermidis*.²³ Chemical modification of poly(ethylene) (PE) with sulfonate groups was found to be effective due to the combination of introduced hydrophilicity and negative charge.²⁴

Chemical attachment of polymers provides improved adhesion and thinner layers, but usually requires an elaborate and therefore costly procedure. A polymer can either be chemically attached to surfaces by one of its ends (grafting), or in cases of polyfunctional macromolecules, via several groups in the polymer chain. The first step of polymeric modification of surfaces is in most cases the activation or functionalization of the material that is to be coated, which is commonly achieved by chemical modification, plasma treatment or exposure to radiation. For instance Vaudaux et al.²⁵ describe a system, where PHEMA was graft polymerized onto a polyester-polyurethane material activated by radiation. The water-soluble polymer PEG was attached by one hydroxyl end-group, resulting in surfaces that show excellent microbe-repelling properties.^{16,19} This effect can even be improved by modifying the remaining PEG

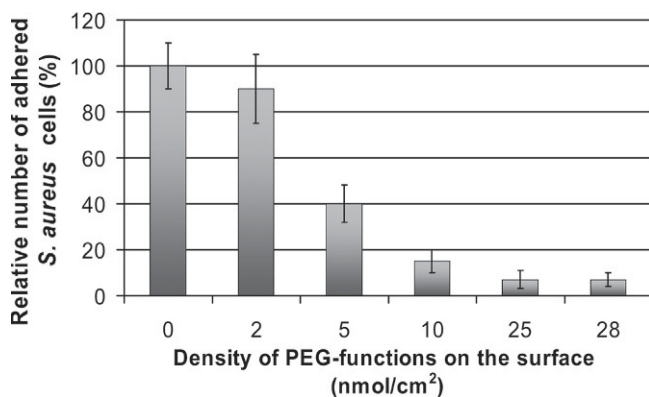


FIGURE 18.5 Adhesion of *S. aureus* on PEG-modified collagen in dependence on the PEG grafting density.

OH-group with a negative charge. As described by Han et al.²⁶ the adhesion of *S. epidermidis* could be lowered more than fivefold compared to the unmodified PEG-grafted surface. As seen in Figure 18.5, the increase in the density of PEG-groups on a PEG-modified collagen-coated surface lowers the adhesion of *S. aureus* cells up to 93% compared to the unmodified collagen surface.

Another way of modifying surfaces with hydrogels, consists of blending together of water-insoluble and water-soluble polymers. In an appropriate system, the polymer is immobilized on the surface and does not migrate into the surrounding solution. For example, blending of PEG and PET leads to a surface which reduced adhesion of *S. aureus* by a factor of up to 20 under various conditions.²⁷ Similar results were obtained with surfaces modified with pluronics, which also contain PEG.²⁸

The problem of hydrogel-like coatings is that they always need to be wetted prior to use in order to be effective and therefore will not work for airborne microbes. It was found in our own recent unpublished studies that a glass surface coated with a polyacrylic acid did not alter the adhesion of *S. aureus* cells from their suspension compared to the unmodified control when applied to the bacterial suspension in dry state. If this surface is wetted in sterile buffer for only one second and then immersed into the bacterial suspension, the adhesion drops by a factor of 20.

18.8.2 Ultrahydrophobic Coatings

The surface energy of surfaces plays an important role in microbial adhesion. Depending on their adopted environment, microbes themselves can have hydrophilic or hydrophobic surfaces.²⁹ Consequently, the microbial cells adhere preferably to compatible surfaces, i.e., hydrophobic cells adhered better to hydrophobic surfaces. In spite of this rather uncontrollable phenomenon it was found that

rendering the surface energy to extremely low (ultrahydrophobic) values could prevent microbial adhesion.

The approach of low-energy surfaces for repellence of microbes can be realized for almost every material by conventional dip, flow or spin coating with ultrahydrophobic polymers, such as poly(methylpropenoxy-fluoroalkylsiloxane) or poly(perfluoroacrylate),³⁰ or chemically modifying surfaces with protic functions (such as hydroxyl groups), e.g., glass, metals or ceramics, with fluorosilane, such as poly(fluorotrimethoxysilane).³¹ The resultant surfaces show a strong repelling potential for microbes, e.g., for the pathogenic bacterium *Streptococcus mutans*.

It was found recently that it is possible to coat surfaces in a way that they show the so-called lotus effect. This effect is the typical behavior of the lotus plant, whose leaves repel water so strongly that drops bounce off the surface, and roll over it as near perfect spheres. It is caused by a unique nanostructured surface morphology combined with a surface-bond hydrophobic substrate. This effect is achieved for synthetic materials by nanostructuring of a surface, e.g., by plasma treatment followed by chemically modifying it with a fluorosilane to form ultrahydrophobic surfaces. Almost every material can be altered this way. The antimicrobial properties of lotus effect materials are under investigation, but according to previous results with even less hydrophobic substrates microbes should be repelled very efficiently.

18.8.3 Influence of Surface Net Charge on Microbial Adhesion

Charges can be introduced into plastic surfaces by plasma treatment in the presence of oxygen (negative charges) or of ammonia (positive charges), as well as by various coatings with charged polymers. Further treatment of materials with functionalized silanes, e.g., 3-aminopropyl-triethoxysilane, is used to chemically charge surfaces. The effect of charge is dependent on the material and the targeted micro-organism, i.e., the application. It was found that in the case of biofouling caused by *P. aeruginosa* on water processing membranes, the surface charge, whether positive or negative, must be minimized to obtain minimal biofouling.³² Other authors report that increasing negative charge reduced bacterial adhesion,³³ while increasing positive charge enhanced adhesion but inhibited growth.³⁴

18.8.4 Proteins

It has been found that adhesion of plasma proteins, e.g., albumin, can significantly lower bacterial adhesion. Most bacterial, but also a number of fungal cells, exhibit a net negative surface charge. Albumin is negatively charged as well and exhibits no specific binding sites to microbial cells. Therefore, a surface coated with albumin electrostatically repels microbes. For instance, it was shown that fluoroethylene-propylene copolymer surfaces immersed for only 5 min in an aqueous 1% albumin solution can significantly lower the adhesion of *Streptococci*³⁵ compared to the

non-coated material. Similar observations were made for numerous materials, such as titanium and silicone^{36,37} investigated with a variety of micro-organisms, such as *S. aureus* and *S. epidermidis*. In our own unpublished experiments the adhesion of *S. aureus* cells to glass could be lowered by 95% just by treatment of the glass with 10% calf serum for 10 min. The albumin adsorption can be enhanced by surface modifications, such as introducing positive charges or attaching alkylated poly(4-vinylpyridine).^{38,39} Such positively charged surfaces can selectively adsorb albumin from mixtures such as blood plasma or serum. In this way, mostly the bacteria repelling protein albumin and not the bacterial adhesion promoting protein fibronectin adhere to the surface.

18.9 MICROBE KILLING OR GROWTH INHIBITING COATINGS

Adhesion of microbes is an active process, controlled by the cell metabolism. This is why the cell must be alive to adhere to a surface. A microbiocidal coating prevents microbial adhesion by killing the microorganism both in contact with and in close proximity to the surface, e.g., by releasing an antimicrobial agent or directly on contact. With this the microbes are no longer able to adhere and form biofilms.

18.9.1 Release Systems

Most microbe-killing drug release coatings are based on slow diffusion of an antimicrobial agent through a material barrier.^{40,41} The coating procedure used is strongly dependent on the stability of the antimicrobial agent. For instance, metal ions such as silver can endure rather drastic conditions and consequently a wide range of coating methods, e.g., powder coating, polymer melts, and electrodeposition. In contrast, the number of coating methods is rather limited for organic drugs, e.g., peptides, antibiotics, and furanones⁴² owing to their instability, leaving only methods that work at low temperatures and other moderate conditions, such as dip, flow or spin coatings. Typical biocides for release systems are shown in Figure 18.6.

Another possibility for antimicrobial drug release is based on the slow hydrolysis of covalent chemical bonds. The drug is thereby either bonded to the surface matrix, or it is incorporated into a degradable (usually polymeric) network. The drug is then released either by degradation of the supporting matrix or by cleaving of a chemical bond to the matrix.⁴³ Examples of release systems are given in Table 18.1.

18.9.1.1 Diffusion Systems

The easiest way to achieve a drug releasing system is to impregnate a material, preferably a swellable plastic, with a microbiocidal drug just by immersing it into an aqueous or organic solvent solution for a certain period of time. The drug

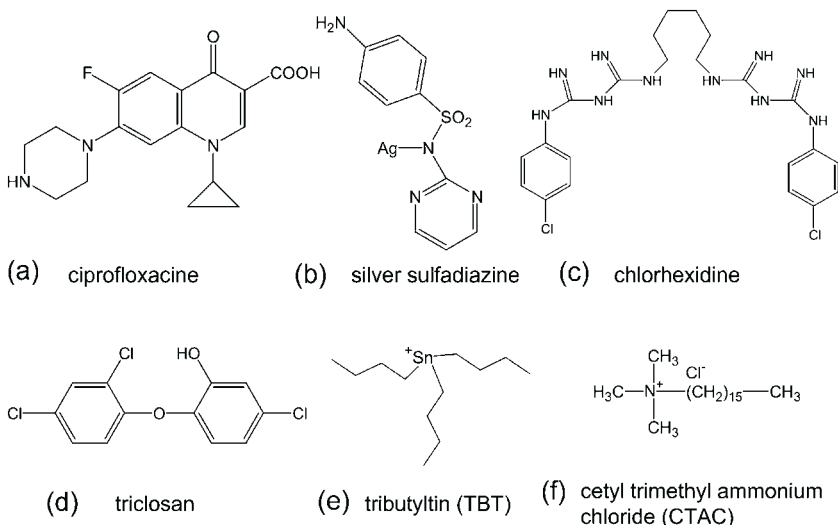


FIGURE 18.6 Structures of typical biocides in release systems. (a) Ciprofloxacin, (b) Silver sulfadiazine, (c) Chlorhexidine, (d) Triclosan, (e) Tributyltin (TBT), and (f) Cetyl trimethyl ammonium chloride (CTAC).

will then tend to migrate into the upper layers of the material, which is swollen to a certain extent depending on the solvent. After removing the latter, the drug is slowly released into an aqueous environment and therefore keeps the material microbiocidal for some time after the treatment. Biological contaminations will be removed during this procedure as well, which is why this method is widely used for extended disinfections of surfaces. Unfortunately, the respective drug is exhausted rather quickly. The release rate is dependent on the matrix properties, such as hydroaffinity, pore size, or swelling ability, as well as on the nature of the agent itself, e.g., size, specific and unspecific interactions with the matrix, and solubility in the respective solvent (Hildebrand parameter). The content of the antimicrobial drug is usually limited to a minor amount so as not to interfere with the material properties, such as hardness, film-forming properties, and adhesion to the coated material. In order to extend the releasing time the following approaches are described.

Increase the amount of drug loading by

- chemical or physico-chemical surface modification prior to impregnation
- coating prior to impregnation
- loading the coating material

Controlling the drug release by

- use of sparingly soluble drugs
- surface coating subsequent to impregnation

TABLE 18.1 Examples of Microbiocidal Coatings that Release Biocides

Material	Surface Treatment	Release Principle	Active Compound	Tested System	Reference
PU	Swelling with antibiotic and the pore-forming agent PEG	Diffusion through pores formed	Ciprofloxacin	<i>E. coli</i>	44
Silicone	Ion beam assisted deposition of Ag	Slow dissolving	silver ions	<i>S. aureus</i> , <i>E. coli</i>	45
PU	Hydrocath coating impregnated with antibiotic	Diffusion	Benzalkonium chloride	<i>S. epidermidis</i>	46
PU	Biodegradable polymer, PU-like	Biodegradation of antimicrobial polymer	Ciprofloxacin	<i>P. aeruginosa</i>	47
Silicone	PEG-Gelatin-hydrogel coating impregnated with antibiotic	Diffusion	Ciprofloxacin	<i>P. aeruginosa</i>	48
Steel and alloplasts	Painting	Cleavage of chemical bond	TBT	<i>Barnacle nauplii</i>	49
PU	Immobilization of antibiotic via 1,6-diisocyanatohexane	Cleavage of chemical bond	Rifampicin	<i>S. aureus</i> , <i>S. epidermidis</i>	43
PU	Swelling with antibiotic solution	Diffusion	Gendine	<i>S. aureus</i> , <i>P. aeruginosa</i> , <i>C. parapsilosis</i>	50
Titanium	Anodic oxidation followed by impregnation with antibiotic	Diffusion	Gentamycin	<i>E. coli</i>	51
Wood	Painting	Diffusion	Borate salts	Bacteria and mildew growth	52

Alloplasts	Chemical modification with polyguanidine and loading it with silver ions	Diffusion exclusively into adhered microbes	Silver ions	<i>E. coli</i> , <i>S. aureus</i> , <i>P. aeruginosa</i>	53
Glass, textile fibers	PU coating with n-halamine copolymers	Diffusion	Chlorine	<i>E. coli</i>	54
Steel, aluminum	Impregnated with silver/zeolite	Diffusion from complex	Silver ions	<i>E. coli</i> , <i>S. aureus</i>	36
Vinylidene chloride	Impregnation	Diffusion	K+sorbate, Na+benzoate	<i>Listeria monocytogenes</i>	55
PE	Co-extrusion	Diffusion	Hexamethylenetetramin	<i>Lactobacillus</i>	56
PE	Multilayer co-extrusion, sol-coating	Diffusion	Grapefruit seed extract	<i>E. coli</i> , <i>S. aureus</i> , <i>Bacillus subtilis</i>	57
Plastics	Co-extrusion Coating	Diffusion	Triclosan	Numerous bacteria and fungi <i>E. coli</i> , <i>S. aureus</i>	58 59
Ceramics	Coating by firing	Diffusion	Zn, Ag, Cu, org compounds	<i>Legionella</i> , <i>Vibrio</i> , <i>Salmonella</i> , <i>E. coli</i>	60
Polystyrene	dip coating with <i>Pseudoalteromonas tunicata</i> film in hydrogel	Diffusion and production	Bacterial biocides	<i>Barnacle nauplii</i> , <i>E. coli</i>	61
Silicone	Xerogel coating	Diffusion	none	Tested on rats	62

- formation of complexes between the drug and the coating matrix
- use of additives in the matrix, e.g., pore-forming agents
- use of degradable matrices or labile chemical bonds between matrix and drug

18.9.1.1.1 Chemical or Physico-Chemical Surface Modification prior to Impregnation

Important factors of absorbing drugs in matrices are the specific types of chemical interactions (e.g., electrostatic, van der Waals forces, hydrophilic/hydrophobic interactions, and hydrogen bond forming potential). It was shown that a significantly larger amount of antimicrobial drug is absorbed by modifying the surface material with the opposite charge. This effect was achieved for instance for ethylene–vinyl alcohol copolymers by chemical modification with negatively charged groups, whereby the absorption of the positively charged antibiotic cephalosporin was improved by 50% compared to the unmodified polymers (1.3–1.8 mg/cm²). This leads to an increased duration of the antimicrobial activity from 24 to 72 h.⁶³

Surface modification with the negatively charged molecules, such as multiple charged heparin, also lead to an increased absorption of an antimicrobial drug. For example, polyurethane (PU) was chemically surface-modified with heparin followed by impregnation with either benzalkonium chloride or a chlorhexidine-silver sulfadiazine mixture. Both were active against a wide variety of potential microbial pathogens, including the pathogenic yeast *C. albicans*. This example also shows how much the release time is dependent on the nature of the drug and the support matrix. While the benzalkonium chloride impregnated material lost 50% of its activity within 24 h, the chlorhexidine-silver sulfadiazine system was completely active after the same period of time.⁶⁴

Especially for materials that cannot be swollen, it greatly increases the absorption of biocide when a porous layer is formed by chemical modification of the surface. For example, anodic oxidation of titanium surfaces leads to a negatively charged porous layer with pore sizes from 0.1 to 1.5 μm . Positively charged antibiotics such as ciprofloxacin and gentamycin adsorb to this layer and are slowly released for an effective time of up to 2 weeks.^{51,65} Anodic oxidizing of aluminum and subsequent absorption of the biocide iodine resulted in a similar effect. The coating exhibited a high activity against the bacteria *E. coli* and *S. aureus* as well as several molds.⁶⁶

Alternatively, materials can be coated with various polymers, such as hydrogels, that are able to take up a large amount of the microbiocide. An example of a hydrogel-based release system consists of a surface (silicone) immobilized PEG-gelatin formulation containing liposomes and the antibiotic ciprofloxacin. The released antibiotic showed antimicrobial efficacy against *P. aeruginosa* for more than 7 days.⁴⁸ Investigations of PU materials with the commercially

available coating Hydrocath, impregnated with the quarternary antimicrobial agent benzalkonium chloride, exhibited prevention of microbial colonization with several *S. epidermidis* and *S. aureus* strains. It could be shown that a high initial rate of killing prolonged microbiocidal activity.^{46,67} Rojas et al.⁶⁸ describe a hydrogel coating for polyurethane, which is applied by simple immersion of PU into a suspension of a prepolymer and a bioactive human immunoglobulin (IgG) in isopropanol. The release of the IgG was 50 $\mu\text{g}/\text{cm}^2$ within several hours exhibiting a strong reduction of *E. coli* adherence.

18.9.1.1.2 Loading the Coating Material

The largest amount of an antimicrobial drug can be impregnated when the coating material is loaded with the drug prior to coating. This consequently increases the effective releasing period compared to other impregnation methods and can be controlled by the thickness of the coating. It can be achieved by swelling the material in a solution of the drug in an organic solvent by mixing both, or in case of polymeric coating materials, by co-extrusion.

A well investigated and also industrially used releasing system is triclosan incorporated into polymers, such as polypropylene, polyethylene, and nylon.^{41,58} The systems slowly release the biocide over a period of years. The incorporation is achieved by co-extruding the polymer and the biocide. It can either be used as whole material or as coating for basically any material. The only downside is the constant release that might be toxic to the applicant and at least will build up microbial resistance, which is especially dangerous for products that are in constant contact with the product. Another example is the co-extrusion of grapefruit seed extract with polyethylene. Films formed from this material were found to reduce growth of bacteria, such as *E. coli*, *S. aureus*, and *Bacillus subtilis*.⁵⁷ For instance, silicone materials loaded by swelling them in a 0.2% solution of rifampicin and clindamycin HCl in chloroform for 1 h showed an effective antimicrobial release into an inoculated brain-heart-infusion for more than 56 days, a time during which no colonization with 17 different staphylococci strains occurred.⁶⁹

An example of a coating loadable with quaternary ammonium ions from its aqueous suspension is based on amphiphilic co-networks made of polydimethylsiloxane and a copolymer of acrylic acid and 2-hydroxyethyl acrylic acid.⁷⁰ The coating can be loaded repeatedly and is antimicrobial for at least 3 weeks each time.

Besides metal ions and low molecular weight drugs, antimicrobial polymers, such as polymeric phosphonium salts, were also found to be effective against micro-organisms when used in a release system, e.g., when incorporated into the common plastic material PET.⁷¹ Another approach is based on coating a variety of materials with a sulfonated triblock copolymer that allows subsequent loading with biocides.⁷²

18.9.1.1.3 Use of Sparingly Soluble Drugs

Surface coatings with sparingly soluble antimicrobials do not show the common outburst of release systems, allow a greater loading and a slower release, which altogether affords the release system to be effective for long periods of time. One example of such a system is silicone, either dip coated with micronized silver oxide or impregnated with it by ion beam deposition. Both coatings resisted *S. aureus* infection for 9 months in vitro.⁷³

18.9.1.1.4 Coating After Impregnation

In order to reduce the releasing rate of an antimicrobial agent, the impregnated material can be coated with a polymer film that exhibits a low-diffusion coefficient for the drug, i.e., the drug release will be slowed down, extending the release time. For example, the release of the antimicrobial drugs silver nitrate, benzylpenicillin, and tetracycline from the biodegradable bulk matrix poly(lactic-co-glycolic acid) were successfully slowed by coating the material with ethylcellulose subsequent impregnation.⁷⁴ The biocides are released in vitro for more than 4 weeks with zero-order kinetics. A system based on the same principle that led to the longest effective antibacterial action so far is based on a sandwich structure of an antibiotic between two silicone sheaths. It was found to be effective against *Staphylococci* for more than 3 months and even after 325 days it still exhibited antimicrobial properties.⁷⁵

18.9.1.1.5 Complexes

A very effective way of slowing down the migration of antimicrobial drugs from impregnated matrices is to form complexes between the support matrix of the surface and the drug. Owing to the strong binding constant, the drug will be released in a slow and controlled manner. One recently developed and already commercially available system (Zeomic®) consists of silver ions, a very effective biocide against the majority of micro-organisms, encapsulated in an inorganic host structure, a zeolite.⁷⁶ The complex can be used as a release system for several applications described below, because it can be incorporated into or used as coating for many materials, such as polymers, ceramics, and metals. One product name is AgION®.

18.9.1.1.6 Additives

Especially for impregnation of hydrophobic materials, it often occurs that the impregnated material does not release the loaded biocide fast enough in order to reach the critical environmental concentration. To address this problem, the drugs are co-impregnated with so-called pore-formed agents, which are basically water-soluble compounds that release quickly and leave channels and pores behind. The incorporated drug is then able to slowly migrate through these pores and reach an effective antimicrobial concentration on the surface

of the material. One example of such a release coating is a system where PU is loaded with ciprofloxacin, a wide-range antibiotic, in the presence of poly(ethylene glycol) as pore-forming agent.⁴⁴ This formulation coated onto a surface was effective against several bacterial strains for more than 5 days.

18.9.1.1.7 Release by Hydrolyzing Labile Bonds

A rather controlled release of highly loaded drugs can be achieved by immobilizing the drug either directly onto a surface or into a surface coating material via a labile, hydrolyzable bond. The surface will then be stable in a non-aqueous system and only release the drug slowly in an aqueous and therefore potentially infectious environment. One example is the immobilization of the antibiotic ciprofloxacin on PU and silicone via a hydrolyzable urethane bond.⁴³ The release of this drug was extended for 6 days and proved to be effective against various *Staphylococci* strains. The best investigated coatings that release biocides by hydrolyzing bonds are antifouling paints, which are described in detail below.

18.9.1.1.8 Biodegradable Systems

Numerous biodegradable polymer systems have been developed in order to equip materials with antimicrobial coatings. Poly(lactic acid) was also successfully used to release gentamycin.⁷⁷ The antibacterial protein lysozyme was incorporated into a cross-linked gelatine matrix, which released the enzyme slowly over 50 h.⁷⁸

A system that releases an antimicrobial drug only in the presence of a bacterial infection was developed by Tanahira et al.⁷⁹ It is based on the fact that in case of a wound infection the concentration of the blood coagulation enzyme thrombin increases dramatically. The developed system contains the antibiotic gentamin incorporated into a poly (vinyl alcohol) matrix, which is cross-linked by means of thrombindegradable peptide linkers.⁸⁰ It could be shown that gentamin is only released when an infection with, e.g., *P. aeruginosa*, occurs (see Figure 18.7). A disadvantage of this system is that in vivo the bacteria might form a biofilm, which is usually more resistant than suspended bacteria,⁸¹ before the antibiotic is released. Further, it can be used only once, because the thrombin will degrade the whole network.

Other than being incorporated into a matrix, the antimicrobial can be part of the coating material matrix. For example, a network composed of polycaprolactone diol, the antibiotic, ciprofloxacin, and the cross-linking agent 1,6-hexane diisocyanate, shows multiple biodegradation releasing the antibiotic slowly over a period of 10 days.⁴⁷

18.9.1.1.9 Biological Release Systems

An alternative antimicrobial release coating uses bacteria that produce a wide-range microbiocide incorporated into a coating matrix. The advantage of such a

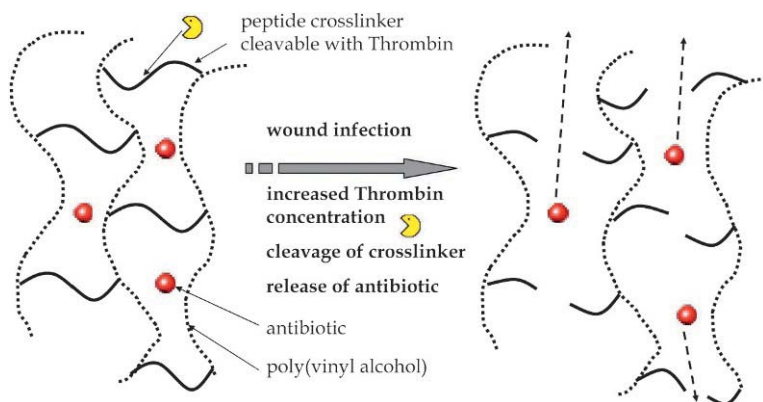


FIGURE 18.7 Concept of an antimicrobial release on demand system according to Tanahira et al.⁷⁹

system is that the system will not exhaust until the bacteria die, which, depending on the environment, could take a long time, owing to their strong viability. So far a system is described, where the marine bacterium *Pseudoalteromonas tunicata* is incorporated into a polyvinylalcohol-hydrogel. The loaded gel releases antifoulants produced by the bacteria and is efficient against *Barnacle nauplii* for over 2 weeks and against *E. coli* for up to 2 months.⁶¹

18.9.1.1.10 Coatings that Release on Microbial Contact

A recently described coating is based on the complexation of silver ions with a covalently attached polymer containing guanidine functions. The ions do not release into the environment but into bacterial cells, which are directly in contact with the surface. This system, called Surfacine, could be applied to numerous alloplastics.⁵³ It is effective against a wide range of bacteria without being released into the surrounding solution.

18.9.2 Contact Active Antimicrobial Surfaces

Surfaces that keep themselves clean by killing adhered microbes on contact without releasing a possibly toxic agent are most desirable because they are environmentally friendly and reduce the risk of developing microbial resistance, a major problem in modern medicine.⁸² Although the principles of contact killing surfaces are not fully understood the methods can be classified as functioning either by chemical or by physico-chemical action. Typical examples for contact active coatings are given in Table 18.2.

TABLE 18.2 Examples of Microbiocidal Coatings that Kill Microbes on Contact

Material	Surface Treatment	Active Compound	Tested System	Reference
Polystyrene	Chemical modification	Peptides	<i>E. coli</i> , <i>S. aureus</i> , and others	83
Steel	Dip coating (microemulsion)	TiO ₂	<i>Bacillus Pumilus</i>	84
Glass	Nanocomposite coatings	TiO ₂	<i>E. coli</i> , <i>B. megaterium</i>	85
Silicone, rubber	Chemical modification	Quarternary aminosilanes	<i>S. aureus</i> , <i>S. epidermidis</i> , <i>E. coli</i> , <i>P. aeruginosa</i>	86
Polystyrene	Blending with chemically modified polymer	Iodonated fibers	<i>E. coli</i> , <i>C. albicans</i>	87
Cellulose	Chemical modification	Aldehyde groups	<i>E. coli</i>	88
Zn and Zn-coated metals	Oxidation of electroplated Zn coatings	ZnO	<i>Aspergillus strains</i> , <i>Penicillinium</i> , <i>C. albicans</i>	89
Cellulose	Ultrasound assisted coating	ZnO	<i>E. coli</i>	90
PET	Multilayer with heparin	Chitosan	<i>E. coli</i>	91
PU	Chemical modification	Quaternary ammonium functions	<i>S. aureus</i> , <i>E. coli</i>	92
Glass, plastics	Chemical modification Aqueous coating	Hexyl-PVP	<i>P. aeruginosa</i> , <i>S. Epidermidis</i> ,	9
		Methyl-PVP	<i>E. coli</i> ,	10
			<i>S. aureus</i>	93
Polypropylene	Chemical modification	Phosphonium-polymer	<i>S. aureus</i> , <i>E. coli</i>	94
Nylon	Chemical vapor deposition	Tert. amine polymer	<i>E. coli</i>	95
Glass	Coating	Alkylated PEI	<i>E. coli</i> , <i>S. aureus</i>	96
Cellulose	Chemical modification	Quaternary aminoacrylates	<i>E. coli</i> , <i>B. subtilis</i>	97

18.9.2.1 Chemical Action

The principle of chemical contact killing surfaces is based on a chemical reaction, such as oxidation or hydrolysis, between the surface, the environment and the adhered microbes. The microbial cell walls or membranes are thereby disrupted leading to their death. A recently developed system is based on coatings containing photochemically active nanoparticles of TiO_2 . These coatings are activated by light and thereby form active oxygen species, e.g., hydroxyl radicals, which have a short half life and therefore exist only in close proximity to the surface.⁹⁸ These species are then capable of killing microbes, for instance by oxidizing the phospholipids of the microbial cell membrane.⁹⁹ The killing efficiency is more than 99% tested for numerous microbes, such as *S. aureus*, *E. coli*, and *C. albicans*.¹⁰⁰ The only downside of TiO_2 , which can be applied by sol-gel processing, sputtering, spraying, as paint additives, or in crystallized coatings (e.g., hydroxyl apatite), is that it requires UV-light to be effective. Recent developments have shown that doping TiO_2 with silver ions results in materials that are photocatalytically active in room light.¹⁰¹ Doping with vanadium ions and gold resulted in daylight active antimicrobial coatings as well.⁸⁵ Such coatings are already used for instance in wall paint for the home.

The photocatalyst WO_3 was found to be effective against microbes and algae with the same efficacy as TiO_2 ,⁷⁸ but because of its high price and toxicity it is not used for industrial applications. Another oxidizing coating is ZnO , which can be generated by anodic oxidation of electroplated Zn coatings. It was tested successfully to be effective against numerous pathogenic yeasts and fungi.⁸⁹ The principle of this likely oxidizing coating, as well as coatings based on toxic, surface-immobilized biocides, such as organo-tin and mercury compounds, is not entirely clear.^{103–105} Although the latter modifications have proven to be very effective against bacteria, fungi, and yeast in laboratory tests, they are not usable for most applications, because of their high environmental toxicity which has also prohibited the use of these biocides.

18.9.2.2 Physicochemical Action

The functional principle of physicochemically active coatings is based on a physical or physicochemical interaction between molecules covalently (i.e., via a stable chemical bond) attached to a surface and microbial cells adhered to it. Due to these interactions the surface will either inhibit the proliferation (growth) of the adhered microbes or kill them mostly by disrupting the microbial cell walls and membranes. Such systems are very effective and theoretically are not exhausting environments with rather low contamination (up to 10^6 cells per ml). This is because microbial cells are killed upon adhesion and then detach from the surface. In this way the surface will always be effective. If the microbial cell concentration is too high the antimicrobial surface will become saturated with cells. A second cell layer, which would then have no contact with the deadly surface, might occur and grow a biofilm.

One example of a growth inhibiting coating is the introduction of a highly positively charged surface, either by chemical modification of plastics (introduction of amino groups),⁸⁶ or by coating of a surface with a positively charged polymer, such as chitin or chitosan (deacetylated chitin, containing amino groups, dip coated from the concentrated aqueous solution).¹⁰⁶ In all these cases, microbes, such as *S. aureus* or *P. aeruginosa*, are adhered in viable state but their proliferation is hindered. Thus, they exhibit less potential to grow biofilms (Figure 18.8).

In contrast, it was found that most highly effective low-molecular weight and polymeric antimicrobials lose the majority of their efficiency when incorporated into or immobilized onto material surfaces.^{107,108} It is reasonable that if these drugs cannot reach their target, which is usually the cytoplasmic membrane or the cytoplasm itself, they will not be effective. In particular, when such polymeric drugs (Figure 18.9) are immobilized on surfaces by grafting (fixation to one end of the polymer), the long chain can still migrate into

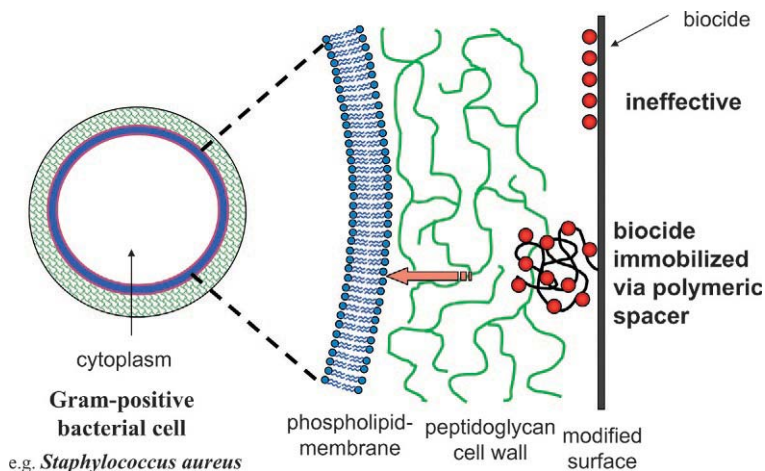


FIGURE 18.8 Concept of contact-active antimicrobial coatings with immobilized biocides.

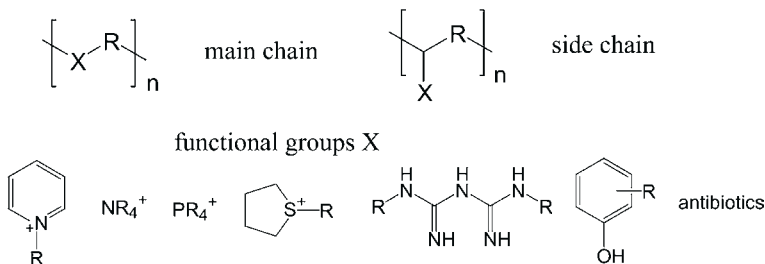


FIGURE 18.9 Examples for antimicrobial polymers: main chain, side chain, functional groups X, and antibiotics.

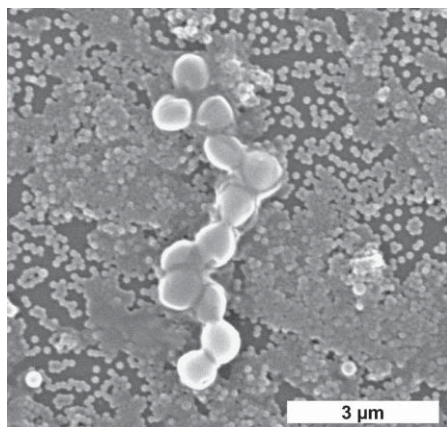


FIGURE 18.10 SEM image of *S. aureus* cells killed on a biocidal nanoparticle coating.

the microbial cell and kill it (Figure 18.8). It was found that poly(4-vinyl-N-alkylpyridinium) salts (alkyl-PVP) grafted on glass, kill bacteria most efficiently on contact.⁹ Bacteria such as *S. aureus*, *S. epidermidis*, *P. aeruginosa*, and *E. coli* were killed up to 99% in either airborne or waterborne states. By using combustion chemical vapor deposition, which is able to coat almost every material with an ultrathin active SiO₂ layer,¹⁰⁹ the alkyl-PVP modification could be performed on numerous plastics, such as polyethylene, polypropylene, and poly(ethylene terephthalate), all with similar antibacterial effects.¹⁰

As a result, the density of the pyridinium groups, which is a measure of the chain length of the polymer, is only effective above 3 nmol/cm². Another of these rather elaborate chemical grafting techniques led to contact-active antimicrobial modifications of cellulose with quaternary poly (acrylates)⁹⁷ and of glass with quaternized poly (ethylene imine) (PEI).¹¹⁰ In order to obtain the same type of coating without the elaborate chemical modification, water-insoluble PEI-type polymers were developed that can be applied by simple coating with solutions of the polymer in organic solvents.⁹⁶ Even more environmentally friendly are recently developed aqueous suspensions of polymer nanoparticles modified with the antimicrobial polymer methyl-PVP.⁹³ Using these suspensions allows the application of contact-active coatings to any material by treating it with an aqueous suspension. Figure 18.10 shows an SEM image of killed *S. aureus* cells on such a coating.

18.10 METAL COATINGS

Numerous transition metals are known to be antimicrobial in their elemental form. The best investigated one is silver, which was used unconsciously to prevent infections for more than 2,000 years in the form of silver cutlery or dishes. For example, in the army of Alexander the Great (356–323 BC) the death rate of

the common soldiers was three times higher than that of the officers; presumably, the officers could afford to drink from silver cups. Nowadays, we know that silver cups are inherently antimicrobial, and therefore bore fewer germs than the leather cups used by the common soldiers.

The functional principle of transition metal coatings is still not clear. One opinion is that the metal layers slowly release ions or nanoparticles,¹¹¹ which penetrate micro-organisms on contact or in the surrounding solution and kill them there. On the other hand, it is discussed that the metal layer itself catalyzes the formation of active oxygen species such as hydroxyl radicals or hydrogen peroxide, which then destroy the surrounding microbes.¹¹² Therefore, metal coatings cannot be clearly classified as either releasing or contact active coating. However, the application of transition metal layers is possible on almost every material, varying from plastics to metals and ceramics to natural products. Depending on the material the metal can be deposited by electroplating, electroless, or by chemical deposition, and sputtering.

The most effective antimicrobial metal layer used so far consists of silver. Materials, such as silicone, Latex, Teflon,¹¹³ PU, polyesters,¹¹⁴ nylon,¹¹⁵ textile fibers,¹¹⁶ and ceramics¹¹⁷ are silver coated using magnetron sputtering,¹¹⁸ ion beam sputtering,¹¹⁹ or electroless deposition by reducing AgNO_3 in aqueous solution with a reducing agent such as formaldehyde¹²⁰ in the presence of the target substrate. These coatings proved to be effective against a variety of infectious Gram-positive (e.g., *S. aureus*, *S. epidermidis*) and Gram-negative (e.g., *E. coli* and *P. aeruginosa*) bacteria¹²¹ and were successfully tested in practice.¹²² As an alternative to pure silver coatings, materials can be modified with silver nanoparticles, e.g., by dip coating surfaces with polymer stabilized silver nanoparticles¹²³ or with layer by layer techniques.¹²⁴ Although many materials have been equipped this way, it was found that the antimicrobially active amount of silver is only three times less in the case of silver nanoparticle coatings compared to silver applied by physical vapor deposition.¹²⁵ Metal coating alternatives to silver, such as copper, chromium, platinum, and various alloys of Co, Ni, and Cu, were found to be less effective.^{120,126,127}

The performance of metal coatings can be greatly enhanced by applying low voltage electrical current (0.4–400 μA) to them.¹²⁸ The bactericidal effect of electrical current, which has been proven for numerous bacterial strains, is considered to work by either producing biocidal hydrogen peroxide,¹²⁹ joule-heat,¹³⁰ or by enhanced release of biocidal metal ions, e.g., silver ions.¹³¹

18.11 ANTIFOULING PAINTS

The best investigated, longest known, and most industrially applicable coatings for preventing biological contamination are the so-called antifouling paints. Various principles were applied for achieving such paints. Almost all are based on the release of biocides, which are simply added to the formulation as

drug-filled microbeads, or they are immobilized onto the coating material via a hydrolyzable bond (for reviews see Refs. 132–134).

The most common and effective antifouling coating is based on organotin compounds [e.g., tributyltin (TBT)], which are covalently bound to a polymer backbone, e.g., poly (methacrylic acid). The concept is illustrated in Figure 18.11. When exposed to seawater the chemical bond to the polymer backbone is hydrolyzed, releasing the biocide. Once the majority of the biocide–polymer bonds are cleaved the polymer in the upper layer (note that the active zone is only 10–90 nm in thickness) becomes water soluble, migrates into the surrounding solution, and a new clean surface is exposed.⁴⁹ These so-called self-polishing antifouling paints are effective for up to 3 years against all kinds of maritime micro-organisms and also macro-organisms such as barnacles, tubeworms, and mussels. Owing to the release principle these paints must be renewed frequently (usually once in 2 years). Unfortunately, the commonly released biocides, such as TBT, proved to be very toxic to the environment. That is why these compounds have been banned worldwide beginning in 2003, which means that almost 60% of all existing biocidal products have been taken off the market at this time. A search for comparably active antifouling paints has been in progress for several years now.

Alternative antifouling coatings based on the same principle as the TBT paints, which use less toxic compounds, such as Cu carbonate, Cu₂O or non-toxic zinc- and silicon-organo compounds, are on the market already.¹³⁵ Although they are yet to be proven effective in practice, high loading with copper compounds and addition of so-called booster biocides, such as Irgarol (a triazine), Diuron (Dichlorophenyl dimethyl-urea), and zinc pyrithione, is required. The environmental toxicity of such paints is not yet clear.

An alternative to these synthetic booster biocides is natural biocides, which can be extracted from numerous maritime organisms, such as Gorgonian coral, eel grass, maritime sponges, and the bacteria living on them.¹³⁶ Development of

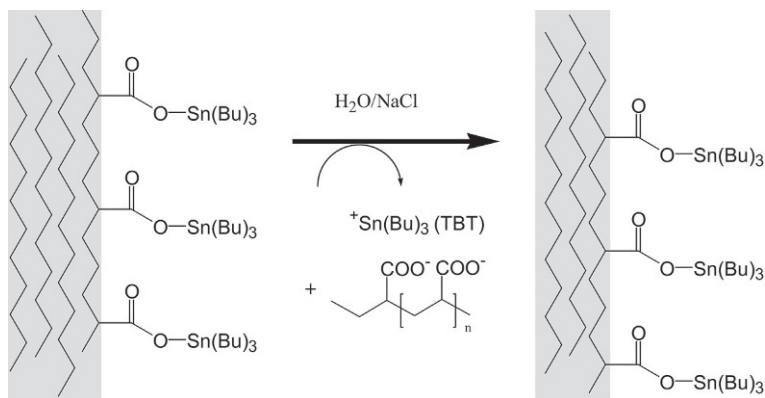


FIGURE 18.11 Concept of a self-polishing antifouling paint.

antifouling paints based on such nontoxic natural biocides is in progress. The main problems here are the limited available quantities, the rather specific activity regarding micro-organism and environmental conditions, and the high costs. Furthermore, these natural organic biocides are less stable than the synthetic ones.

A recently marketed alternative to releasing coatings is an antifouling paint based on photocatalytic TiO_2 , which does not release a biocide, but requires sunlight to be effective (E Paint SN-1).

18.12 ANTIMICROBIAL SURFACES WITH MULTIPLE ACTION

Coatings that have more than one mode of action against microbes have higher performance potential and are the focus of recent research. The first examples of such coatings are the antifouling paints described above, because they repel and kill microbes at the same time. Another approach is based on PEGylated coatings with incorporated silver nanoparticles.¹³⁷ While the PEG repels approaching microbes, the released silver ions kill them. Worley and coworkers have developed a coating that combines two killing mechanisms by creating a polyurethane-coating that contains quaternary ammonium groups for contact-killing and N-halamine groups for releasing chlorine.⁵⁴ A similar concept was realized by incorporating silver nanoparticles into a layer-by-layer coating and finishing it with quaternary ammonium groups.¹³⁸ The combination of silver bromide and polyvinylpyridinium salts resulted in coatings that release silver ions and kill microbes on contact at the same time.¹³⁹

18.13 ANTIVIRAL COATINGS

A special case of biological contamination is virus attachment. Viruses cannot grow on surfaces but are often able to survive and infect organisms on contact. In order to deactivate viruses two approaches are currently being used: releasing an antiviral drug or capturing the virus. An example of a drug-releasing antiviral coating is iodine impregnated polyurethane, which successfully deactivates HIV.¹⁴⁰ In order to capture viruses it is useful to immobilize positively charged polymers or antibodies. For example, the antibody Concanavalin A was immobilized on polystyrene branched with PMMA.¹⁴¹ This coating could capture HIV-1 virions and gp120, rendering them unable to infect cells on contact and therefore having a promising potential to prevent viral transmission. Recently, a polyethylene imine-based polymer was found to inactivate an influenza virus.¹⁴²

18.14 SURFACE CLEANING BY COATING

Most coating methods for rendering surfaces antimicrobial can also be used as cleaning procedures for materials with minor biological contaminations. For example, releasing systems naturally deactivate microbes on the surface that

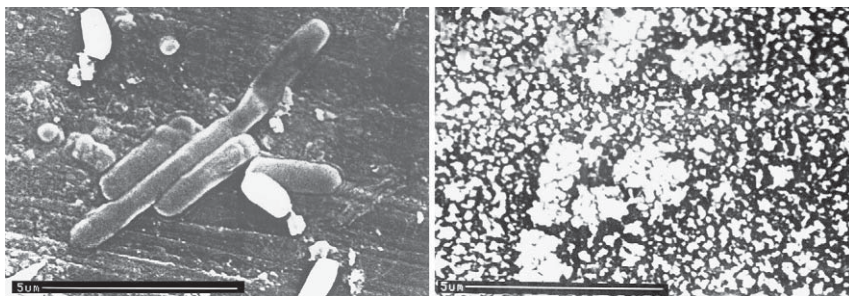


FIGURE 18.12 SEM images of *E. coli* on surfaces before (left) and after (right) coating with SiO_2 .

is to be coated, because they usually release the biocide on both sides of the layer, i.e., there is microbiocidal killing efficacy under the coating as well. This is also true for contact killing surfaces. In the case of surface modifications, the conditions and materials used, such as organic solvents, plasma, glow discharge, heat, etc., are often antimicrobial as well. With few exceptions, e.g., repelling hydrogel coatings, it can be stated that the application procedure and/or the function of antimicrobial and antifouling coatings are not only useful in preventing biological contaminations, but may also inactivate them and/or remove them from surfaces. A recently found coating that kills microbes on surfaces by literally petrifying them is based on the Combustion Chemical Vapor Deposition of SiO_2 , a method to activate surfaces prior to coating. The microbes are thereby coated with a SiO_2 layer, which stops their proliferation and kills them (Figure 18.12).

18.15 APPLICATION EXAMPLES

18.15.1 Biomedical Applications

Rendering catheter surfaces antibacterial has been a field of interest for several decades, because of the crucial need for saving lives and drastically lowering hospital costs caused by catheter-related nosocomial infections that occur in up to 7% (urinary catheters) of all cases depending on the kind of catheter.¹⁴³ Catheters today are usually composed of polyurethane or silicone, which are not naturally microbiocidal. The numerous approaches for rendering catheters antimicrobial, some of which are commercially available, include: coating with hydrogels; impregnating the polymeric materials with antibiotics (most successful with the combinations rifampicin/minocycline as well as chlorhexidine/silver sulphadiazine); and coating them with silver (for recent reviews see 144,145). The most often occurring problem is a lack of long-term efficiency of the impregnations, especially in the case of urinary catheters. So far the longest release period achieved by a system, where the antimicrobial mixture was trapped in a sandwich between two silicone sheaths, afforded an antimicrobial

activity for 1 year.⁷⁵ Even if approved in several clinical trials,¹⁴⁶ antimicrobial catheters are still an active topic of research. A new trend in catheter coatings is the use of silver nanoparticles.¹⁴⁷

Other temporary and permanent implants, such as prosthesis, grafts, stents, shunts, surgical sutures, and cardiovascular devices as well as blood dialysis membranes, are potential targets for nosocomial infections. It is therefore desirable to equip them with antimicrobial properties. Some representative examples are given in [Table 18.3](#) which have been recently reviewed.

TABLE 18.3 Application Examples for Antimicrobial Coatings		
Application	Active Compound(s)	Reference
Catheter	Silver ions	148,149
	Chlorhexidine/silver sulfadiazine	150
	Miconazole/rifampicin	151
	Ciprofloxacin	48
	Silver ions/benzalkonium chloride	152
	Enzyme inhibitors	153
	TiO ₂	154
	PEG-hydrogel	155
Titanium implants	Silver ions	156
	Fluoride complexes	157
	Ciprofloxacin	65
	Chlorhexidine	158
Surgical Sutures	Silver ions	131
	Decamethoxin	159
	Triclosan	160
	Aldehyde groups	88
Wound dressing	Silver sulfadiazine	161
Surgical grafts	Protein-bound nafcillin, cefazolin, cefamandole, silver-pefloxacin	162
Dental Brushes	Chlorhexidine	163
Denture	Histatin	164
Ship hulls	TBT, Cu compounds, booster biocides	49

Continued

TABLE 18.3 Application Examples for Antimicrobial Coatings—cont'd

Application	Active Compound(s)	Reference
Cutlery	Silver ions	165
Textile fibers	Silver ions	116
	Copper ions	166
	Polyhexamethylene guanidine salts, various antimicrobials (e.g. triclosan)	167,168
	Iodinated polystyrene	87
	Quaternary ammonium groups	169
	Tertiary amines	170
	Phosphonium groups	171
	Chitosan	172
Food preservation	Thyme oil, trans-cinnamaldehyde	173
	Cetylpyridinium chloride	174
	Alkylpyridinium iodides	175
Food packaging	Triclosan	58
	Nisin	176
	Hexamethylenetetramine	56
	Grapefruit seed extract	57
	K + sorbate, Na + benzoate	55
	Propyl paraben	177
	Chitin, chitosan	178
Wood protection	Borates	179
Air filters	Decabromodiphenyl oxide, borates	180
	Silver ions	165
	Silk sericin	181
Construction	Fungicides	182
Cutting boards and other plastic products	Triclosan	183
Cable treatment	Co naphthenate	184
Bioglass	Silver ions	185
Glass	TiO ₂ /Fe ³⁺	186
Ceramics	TiO ₂	187
Wall paper	TiO ₂	188

Wound dressings are especially susceptible to microbial infections, which is why they are always treated with antimicrobials. A commercially available wound dressing with antimicrobial properties is Xeroform, which consists of fine-mesh gauze impregnated with bismuth tribromophenate. This was successfully tested in a clinical trial for treatment of wound burns and showed no activity against infection for more than 5 days.¹⁸⁹ Another example of a wound dressing is based on a polyelectrolyte complex composed of chitosan and sodium alginate, which was loaded with silver sulfadiazine.¹⁹⁰ Although, silver-based wound dressings have found to be the most effective ones,¹⁹¹ there are still some antibiotic-loaded materials on the market.¹⁹²

A large field of interest is the antimicrobial coating for dental devices. Pin-tract infections, carries and plaque prevention (all caused by bacteria) all fall particularly within the focus of antimicrobial research. Also dental cement, teeth and teeth substitutes have been coated with antimicrobial systems. The most often used disinfectant in this field is chlorhexidine,¹⁹³ but also tetracycline, triclosan, and benzethonium chloride were applied to dental devices by impregnating them with the antimicrobials.

18.15.2 Food Protection

Food poisoning caused by microbial spoilage is a major problem that results every year in millions of infections and in extreme cases death. In order to enhance the storage stability of food, either the food itself or the packaging material is antimicrobially coated. Most of the successfully used systems are based on the antibiotic Nisin,^{174,194} which is a wide-range antibiotic. Lysozyme and Nisin used in combination successfully lower meat spoilage by impregnating it with the antimicrobials.^{195,196} Other agents for coating food or food packaging to prolong shelf time are chitosan,¹⁹⁷ lactoferrin,¹⁹⁸ butyric acid,¹⁹⁹ and triclosan.²⁰⁰

18.15.3 Textiles

Textiles, especially sportswear, underwear, shoes, and hospital clothes, are able to transmit infections, especially fungal ones. It is common to render textile fibers antimicrobial, either by chemical modification, e.g., introduction of quaternary amino groups, phosphonium functions, or iodide, by impregnating them with antimicrobials, e.g., triclosan or silver zeolite (Livefresh N-NEO® fibers, Japan), as well as coating them with silver. Unfortunately, all these modifications seriously increase the textile costs and often change the properties of the fibers, for instance, rendering them less flexible or lessening their waterand oil-repelling properties.

18.15.4 Daily Life Products

Every day products such as cutlery, bathroom equipment, hospital furniture, kitchen equipment, cutting boards, etc. are increasingly being offered with

antimicrobial coatings or impregnations. One example of household antimicrobial products is Triclosan-releasing plastics (e.g., Microban[®], USA) that are used for instance as cutting boards. A recently marketed product is stainless steel cutlery coated with silver/zeolite (AgION[®], USA), which releases silver ions. Air filters are equipped with positively charged polymers to reduce microbial growth. For numerous other products such as cooling towers, water pipes, food vending machines, ice machines, and cables, all of which could be improved with an antimicrobial coating, practical solutions with respect to costs, efficiency, and low toxicity have not yet been found.

18.15.5 Construction and Ships

Mold-growth and mildew in showers, cellar walls, and on wood products are common problems. Also growth of biofilms on ships and harbor equipment require constant care. All these problems are approached with the so-called antifouling paints. Since all of these paints contain biocides, such as TBT or highly concentrated copper compounds, which are greatly effective but also toxic, alternative nontoxic solutions are under development.

18.16 FUTURE DEVELOPMENTS

Further contact-active coatings, which seem to be the best solution so far, will soon enter the marketplace. It has yet to be proven, whether they are still able to function when strongly contaminated. Developments of antimicrobial coatings in the future will be based on the development of novel nontoxic broad-spectral biocides, which are currently under research. A perfect solution would be an intelligent coating system, which can recognize microbes and release such nontoxic, environmentally friendly biocides only in the presence of biological contamination. This way the system will be long lasting with no side effects.

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